Brandon Sweeting

NSF MPS-Ascend Postdoctoral Researcher, Mathematics

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Research Interests

· Harmonic Analysis · Wavelet Analysis · Dyadic Analysis · Signal Processing · Operator Theory · Singular Integral Operators · Bergman Projection · Bellman Functions · Weighted Theory

Appointments

NSF MPS-Ascend Postdoctoral Research Fellow — 2024 – Present
Washington University in St. Louis — St. Louis, MO

Postdoctoral Researcher — 2021 – 2024
The University of Alabama — Tuscaloosa, AL

Education

PhD in Mathematics — 2014 – 2021
University of Cincinnati — Cincinnati, OH
Dissertation Advisor: Leonid Slavin
Dissertation: *Novel Bellman Estimates for A_p Weights*

BA in Mathematics — 2011 – 2014
University of South Florida — Tampa, FL
Summa Cum Laude

Grants

National Science Foundation (MPS-Ascend) – DMS #2402316 — 2024 – 2027
Harmonic Analysis Techniques in the Study of Operators in Several Complex Variables — $300,000

Publications

1. *On those weights satisfying a weak-type inequality for the maximal operator*
   B. Sweeting, Proc. Amer. Math. Soc. (2026) arXiv:2410.04031

2. *Weighted weak-type bounds for multilinear singular integrals*
   Z. Nieraeth, C. B. Stockdale, B. Sweeting, J. Geom. Anal. (2026) arXiv:2401.15725

3. *Weighted weak-type inequalities for maximal operators and singular integrals*
   D. Cruz-Uribe, B. Sweeting, Rev. Mat. Complut. (2024) arXiv:2311.00828

4. *Weak-type estimates for the Bergman projection on planar domains*
   A. W. Green, C. B. Stockdale, B. Sweeting, N. Wagner [Submitted] (2026) arXiv:2607.09642

5. *Two-weight multiplier weak-type inequalities for maximal operators and singular integrals*
   D. Cruz-Uribe, K. Moen, B. Sweeting [In Preparation]

6. *On the $T(b)$ theorem in spaces of homogeneous type*
   B. Sweeting, B. Wick [In Preparation]

7. *A $T(b)$ theorem for singular integral operators in the Dunkl setting*
   F. D'Emilio, J. Li, B. Sweeting, B. Wick [In Preparation]

8. *Sharp dyadic estimates for A_2 weights via non-infinitesimal Bellman majorants*
   L. Slavin, B. Sweeting [In Preparation]

Conference Talks

AMS 2027 Spring Southeastern Sectional Meeting — 03/20/27
Special Session on Recent developments in harmonic analysis and PDE
University of Alabama at Birmingham, Invited Talk, 20 Minutes

AMS 2026 Fall Central Sectional Meeting — 10/24/26
Special Session on Harmonic Analysis and Intersections Therewithin
University of Minnesota-Duluth, Invited Talk, 20 Minutes

AMS 2026 Fall Southeastern Sectional Meeting — 10/11/26
Special Session on New Developments in PDEs, Harmonic Analysis, and Fractal Geometry
Kennesaw State University, Invited Talk, 20 Minutes

AMS 2026 Fall Southeastern Sectional Meeting — 10/11/26
Special Session on New Developments in Discrete Harmonic Analysis and Applications
Kennesaw State University, Invited Talk, 20 Minutes

AMS 2026 Fall Southeastern Sectional Meeting — 10/10/26
Special Session on Advances in Harmonic Analysis
Kennesaw State University, Invited Talk, 20 Minutes

AMS 2026 Fall Eastern Sectional Meeting — 10/03/26
Special Session on Harmonic Analysis and PDEs
George Washington University, Invited Talk, 20 Minutes

21st Prairie Analysis Seminar — 11/08/25
Kansas State University, Contributed Talk, 20 Minutes

AMS 2025 Fall Central Sectional Meeting — 10/18/25
Special Session on Real and Complex Function Theory
Saint Louis University, Invited Talk, 20 Minutes

14th Ohio River Analysis Meeting — 03/29/25
University of Cincinnati, Contributed Talk, 20 Minutes

Joint Mathematics Meetings 2025 — 01/09/25
AMS Special Session on New Faces in Operator Theory
Seattle, WA, Invited Talk, 20 Minutes

Operators on Analytic Function Spaces — 12/03/24
CIRM (Marseille), Contributed Talk, 20 Minutes

Virginia Operator Theory and Complex Analysis Meeting 10/26/24
University of Richmond, Invited Talk, 50 Minutes

AMS 2024 Fall Eastern Sectional Meeting 10/19/24
Special Session on Recent Advances in Harmonic Analysis
University at Albany, Invited Talk, 20 Minutes

AMS 2024 Spring Central Sectional Meeting 04/20/24
Special Session on Recent Developments in Harmonic Analysis
University of Wisconsin-Milwaukee, Invited Talk, 20 Minutes

AMS 2024 Spring Southeastern Sectional Meeting 03/23/24
Special Session on Geometric Measure Theory and Partial Differential Equations
Florida State University, Invited Talk, 20 Minutes

13th Ohio River Analysis Meeting 03/16/24
University of Kentucky, Contributed Talk, 20 Minutes

Joint Mathematics Meetings 2024 01/06/24
AMS Special Session on New Faces in Operator Theory and Function Theory
San Francisco, CA, Invited Talk, 20 Minutes

Workshop in Analysis 12/08/23
Georgia Institute of Technology, Invited Talk, 20 Minutes

19th Prairie Analysis Seminar 11/03/23
Kansas State University, Contributed Talk, 20 Minutes

AMS 2023 Spring Central Sectional Meeting 04/15/23
Special Session on Inequalities in Harmonic Analysis
University of Cincinnati, Invited Talk, 20 Minutes

12th Ohio River Analysis Meeting 03/18/23
University of Cincinnati, Contributed Talk, 20 Minutes

39th Southeastern Analysis Meeting 03/10/23
Clemson University, Contributed Talk, 20 Minutes

AMS 2021 Fall Southeastern Sectional Meeting 11/20/21
Special Session on Harmonic Analysis and Spectral Theory
University of South Alabama, Invited Talk, 20 Minutes (Virtual)

17th Prairie Analysis Seminar 11/06/21
Kansas State University, Contributed Talk, 20 Minutes (Virtual)

AMS 2021 Spring Central Sectional Meeting 04/17/21
Special Session on Sharp Estimates in Harmonic Analysis
University of Cincinnati, Invited Talk, 20 Minutes (Virtual)

10th Ohio River Analysis Meeting 03/20/21
University of Kentucky, Contributed Talk, 20 Minutes (Virtual)

37th Southeastern Analysis Meeting 03/13/21
University of Florida, Contributed Talk, 20 Minutes (Virtual)

Seminar/Colloquium Talks

Analysis Seminar, Clemson University	10/16/26
Analysis Seminar, Washington University in St. Louis	08/31/26
Analysis Seminar, University of Pittsburgh	04/06/26
Analysis Seminar, Auburn University	12/03/25
Analysis Seminar, Brown University	04/28/25
Analysis Seminar, University of Calgary	04/22/25
Analysis Seminar, Vanderbilt University	11/18/24
Analysis Seminar, University of Missouri	11/12/24
Colloquium, Saint Louis University	09/27/24
Analysis Seminar, Temple University	02/12/24
Analysis Seminar, University of Alabama	10/04/23
Analysis Seminar, Washington University in St. Louis	09/18/23
Analysis Seminar, University of Alabama	09/21/22
Analysis Seminar, Clemson University	09/21/22
Analysis Seminar, University of Alabama	01/29/21

Conference and Workshop Organization

Expanding Pathways in Harmonic Analysis 2027 — TBD
with Alan Chang, Ben Foster, Henri Martikainen, Cristina Pereyra Elodie Pozzi, and Brett D. Wick

48th Summer Symposium in Real Analysis — 06/14/26 - 06/20/26
with Alan Chang, Paul Humke, Cheng-Han Pan, Krystal Taylor Brett D. Wick, and Scott Zimmerman

AMS 2025 Fall Eastern Sectional Meeting — 10/25/25 - 10/26/25
Special Session on Current Developments in Harmonic Analysis
with A. Walton Green, Cody B. Stockdale, and Nathan A. Wagner

AMS 2025 Fall Central Sectional Meeting — 10/18/25 - 10/19/25
Special Session on Harmonic Analysis and Partial Differential Equations
with A. Walton Green, Henri Martikainen, and Brett D. Wick

Expanding Pathways in Harmonic Analysis 2025 — 04/05/25 - 04/06/25
with Elodie Pozzi and Brett D. Wick

Joint Mathematics Meetings 2025 — 01/11/25
Special Session on New Directions in Harmonic Analysis
with A. Walton Green, Cody B. Stockdale, and Nathan A. Wagner

Joint Mathematics Meetings 2025 01/10/25
MAA Project NExT Panel Discussion: Thriving in Academia
with Rachel Bailey and Yesim Demiroglu

Conferences and Workshops Participation

Informal Analysis Seminar 04/26/25 - 04/27/25
Massachusetts Institute of Technology

Harmonic Analysis and Convexity 12/09/24 - 12/13/24
ICERM at Brown University

Field of Dreams Conference 11/08/24 - 11/10/24
AMS National Math Alliance

MAA MathFest 2024 08/07/24 - 08/10/24
Mathematical Association of America

Geometry of Measures and Free Boundaries 07/20/24 - 07/26/24
University of Washington

Harmonic Analysis and Differential Equations 06/23/24 - 06/29/24
Macquarie University

Multivariable Operator Theory Conference 06/10/24 - 06/14/24
Washington University in St. Louis

Extremal Problems in Harmonic Analysis 11/28/22 - 12/02/22
ICERM at Brown University

Workshop on Harmonic Analysis and Related Topics 06/13/22 - 06/17/22
Centre de Recerca Matemàtica

11th International Conference on Harmonic Analysis and PDE 06/06/22 - 06/10/22
Universidad Autónoma de Madrid

CBMS Lecture Series 2020 05/23/22 - 05/27/22
Florida State University

Teaching Experience

Washington University in St. Louis

· Matrix Algebra Spring 2027

· Linear Algebra Fall 2026

· Introduction to Harmonic Analysis Spring 2026

· Introduction to Fourier Series and Integrals Fall 2025

The University of Alabama

· Discrete Mathematics — Spring 2024

· Discrete Mathematics — Fall 2023

· Introduction to Complex Variables — Spring 2023

· Applied Differential Equations I — Fall 2022

· Introduction to Linear Algebra — Spring 2022

· Discrete Mathematics — Fall 2021

University of Cincinnati

· Introduction to Mathematical Reasoning — Summer 2021

· Introduction to Mathematical Reasoning — Spring 2021

· Foundations of Quantitative Reasoning — Fall 2020

· Mathematics of Social Choice — Summer 2020

· Mathematics of Management Science — Spring 2020

· Mathematics of Social Choice — Fall 2019

· Mathematics of Social Choice — Summer 2019

· Mathematics of Management Science — Spring 2019

· Mathematics of Social Choice — Fall 2018

· Mathematics of Management Science — Spring 2018

· Applied Calculus I — Fall 2017

· Mathematics of Social Choice — Spring 2017

· Mathematics of Management Science — Fall 2016

· Applied Calculus II — Summer 2016

University of Cincinnati (TA)

· Calculus II, 3 sections — Spring 2016

· Calculus I, 2 sections — Fall 2015

· Calculus I, 3 sections — Spring 2015

· Calculus II, 3 sections — Fall 2014

Service Experience

- WashU Math Circle Co-Director — Fall 2024 – Present
  Elementary-High School Students with Alan Chang
- WashU Sonia Kovalevsky Math Day Activity Leader — 11/07/26
  To Be Determined
- WashU Sonia Kovalevsky Math Day Activity Leader — 04/25/26
  Factor Game with Georgia Corbett, Jeongwon Lee, and Elaine Yoo
- WashU Sonia Kovalevsky Math Day Activity Leader — 04/12/25
  Polyomino Puzzles and Tic-Tac-Toe with Sydney Mayer
- National Math Alliance Mentored Readings Leader — 02/03/26 – 05/05/26
  Introduction to Harmonic Analysis with Adrian Fry

Honors and Awards

- MAA Project NExT Fellow — Fall 2024 – Present
- Maita Levine Award for Outstanding GA — Spring 2020
- Albert C. Yates Fellow — Fall 2014 – Spring 2021
- King O'Neal Scholar — Spring 2014

Relevant Skills

- *Languages:* English, French (B2), Spanish (B1)
- *Programming Languages:* Python, Java, Swift, Mathematica
- *Programming Libraries/APIs:* TensorFlow, Keras, Scikit-Learn

Professional Activities

- Journal Referee: *Collectanea Mathematica*, *Czechoslovak Mathematical Journal*, *Journal of Mathematical Analysis and Applications*, *Journal of Fourier Analysis and Applications*

Professional Affiliations

- The American Mathematical Society (AMS) — 2014 – present
- Pi Mu Epsilon, Florida Epsilon Chapter (USF) — 2013 – present